Solutions
QA : Algebra - 07

CEX-Q-0230/25

1	-	**2**	2	**3**	4	**4**	3	**5**	2	**6**	-	**7**	4	**8**	1	**9**	3	**10**	2
11	4	**12**	4	**13**	2	**14**	2	**15**	4	**16**	2	**17**	4	**18**	3	**19**	1	**20**	4
21	9	**22**	1	**23**	4	**24**	1	**25**	4										

1. (1) Distance between A and B

$= \sqrt{\{3-(-5)\}^2 + (7-13)^2} = \sqrt{64+36}$

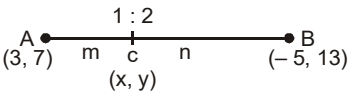
= 10 unit.

(2) The mid-point of AB

$X = \frac{3+(-5)}{2} = -1$

$Y = \frac{7+13}{2} = 10$ which is (–1, 10)

(3)



$X = \frac{1(-5)+2(3)}{1+2} = \frac{1}{3}$

$Y = \frac{1(13)+2(7)}{1+2} = 9$

So, $C \equiv \left(\frac{1}{3}, 9\right)$

(4) Equation of line through AB

$y - 7 = \frac{13-7}{-5-3}(x-3)$

– 8y + 56 = 6x – 18

6x + 8y = 74.

2. 2 It is given that

x + 2y = 9 ...(i)

3x + 5y = 5 ...(ii)

3 × (i) – (ii) gives

y = 22

Therefore, equation (i) gives x = 9 – 44 = –35.

Since the lines are concurrent the above values of x and y should satisfy ax + by = 1

∴ 22b – 35a = 1

3. 4 3x + 4y + 3 = 0 …(i)

3x + 4y + 12 = 0 …(ii)

Put y = 0 in equation (i),

∴ x = –1

∴ (–1, 0) lies on the line (i) perpendicular distance between lines (i) and (ii) is same as perpendicular distance of (–1, 0) from the line (ii)

∴ Required distance = $\left|\frac{3(-1)+4(0)+12}{\sqrt{3^2+4^2}}\right| = \left|\frac{9}{\sqrt{25}}\right| = \frac{9}{5}$.

4. 3 Let ≡ (–a, –b), B ≡ (0, 0), C ≡ (a, b) and D ≡ (a^2, ab)

Here, AB = $\sqrt{a^2+b^2}$

and BC = $\sqrt{a^2+b^2}$

AC = $2\sqrt{a^2+b^2}$

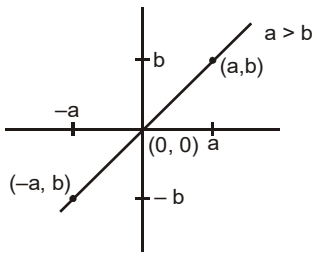
So, A, B and C must be collinear as AB + BC = AC.

⇒ Option (1), (2) and (4) are incorrect.

Hence, option (3) is correct.

Alternative method:

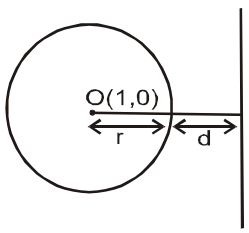
The points A(–a –b), B(0,0) and C(a, b) lie on the same line.



Thus, Option (1), (2) and (4) are incorrect.

Hence, option (3) is correct.

5. 2



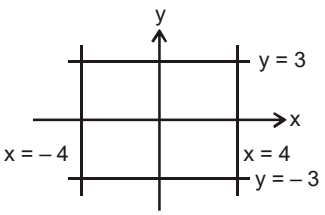
The shortest distance will be the difference between the distance between the centre and the line and the radius.
Centre of the circle is at O(1, 0) and r = 3 units.

$$\Rightarrow r + d = \left|\frac{[8(1) + 6(0) - 48]}{\sqrt{8^2 + 6^2}}\right|$$

$\Rightarrow$ r + d = 4 units

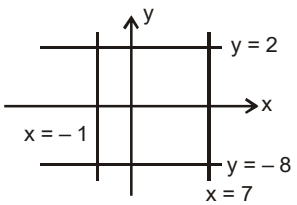
$\Rightarrow$ d = 4 – r = 1 unit.

6. (1) The graph of the given functions is given as
| x | = 4 $\Rightarrow$ x = ± 4
| y | = 3 $\Rightarrow$ y = ± 3



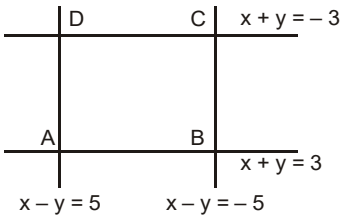
So, required area = 8 × 6 = 48 sq unit.

(2) |x – 3| = 4 $\Rightarrow$ x = 7 and x = – 1,
|y + 3| = 5 $\Rightarrow$ y = 2 and y = – 8



So, required area = 8 × 10 = 80 sq. unit.

(3) Here,
|x + y| = 3 $\Rightarrow$ x + y = ± 3
and |x – y| = 5 $\Rightarrow$ x – y = ± 5
The first two lines are parallel to each other, and other two lines are parallel to each other.
So, the possible graph can be drawn as



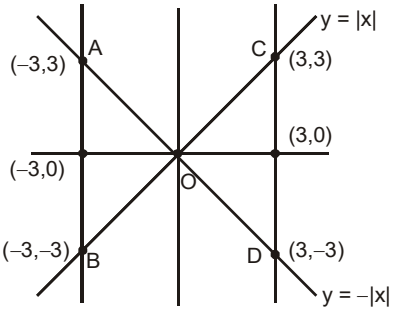
Since, distance between two parallel lines is given by

$$\left|\frac{C_1 - C_2}{\sqrt{a^2 + b^2}}\right|$$

So, AB = $\frac{5-(-5)}{\sqrt{2}} = \frac{10}{\sqrt{2}}$ and BC = $\left|\frac{3-(-3)}{\sqrt{2}}\right| = \frac{6}{\sqrt{2}}$

Thus, area of ABCD = $\frac{10}{\sqrt{2}} \times \frac{6}{\sqrt{2}}$ = 30 sq. unit.

7. 4 Given curves can be drawn as



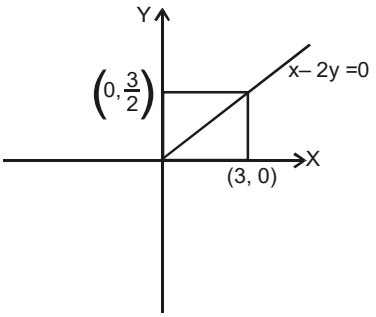
The area generated by the three curves is the sum of area of the triangles OAB and OCD.

$$= 2\left\{\frac{1}{2} \times (6) \times 3\right\} = 18 \text{ square units}$$

8. 1 |x + 2y| + |x – 2y| = 6
On replacing x by –x and y by –y in the above equation, we get the same equation. This implies that the curve is symmetric in the quadrants of X-Y plane
In the first quadrant where x, y > 0
|x + 2y| + |x – 2y| = 6

$$= \begin{cases} (x+2y)+(-x+2y) = 6 \ ; 2y > x \\ (x+2y)+(x-2y) = 6 \ ; 2y < x \end{cases}$$

$$= \begin{cases} y = \frac{3}{2} \ ; \ 2y > x \\ x = 3 \ ; \ 2y < x \end{cases}$$



Area in the first quadrant $= \frac{3}{2} \times 3 = \frac{9}{2}$ square unit

Total area of |x + 2y| + |x – 2y| = 6

$= 4 \times \frac{9}{2} = 18$ square unit.

9. 3 We are to find the solutions for f(f(x)) = 15.
From the graph, f(4) = 15 and f(12) = 15.
The required solutions will be those values of x for which f(x) = 4 and f(x) = 12.
From the graph, the value of function f(x) is 4 at four different values of x, i.e. –8, 1, 7.5 and 10.
The value of the function f(x) is 12 at three different points, i.e. 3, 5 and 11.
Hence, the given equation has 7 solutions.

10. 2 (PL + PN) will be minimum if P lies on LN and (PM + PO) will be minimum if P lies on OM.
$\Rightarrow$ PL + PM + PN + PO will be minimum if P is the point of intersection of the diagonals of quadrilateral LMNO.

LN = $\sqrt{(-5-0)^2 + (0-5)^2} = 5\sqrt{2}$

MO = $\sqrt{(1-(-1)^2 + (-1-5)^2} = 2\sqrt{10}$

$\Rightarrow$ PL + PM + PN + PO = LN + MO = $5\sqrt{2} + 2\sqrt{10}$

11. 4 Since line x = y is perpendicular bisector of AB, B would be a reflection of A.
$\therefore$ Coordinates of B are (–3, –4).
Similarly, BC is bisected by x-axis.
$\therefore$ Coordinates of C will be (–3, 4).

12. 4 Data is insufficient.

13. 2 3x + 4y = 12

X – Intercept:
At y = 0, 3x = 12 or x= 4

Y – Intercept:
At x = 0, 4y = 12 or y = 3

14. 2 ABCD is a rectangle in the clockwise direction. The coordinates of A are (1, 3) and of C are (5, 1) coordinates of vertices B and D satisfy y = 2x + c. Let B be (x_1, y_1) and D be (x_2, y_2).
As diagonals bisect each other, mid-point of AC and BD are same.

$$\Rightarrow (3, 2) = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right) \quad \ldots(1)$$

$\Rightarrow x_1 + x_2 = 6$ and $y_1 + y_2 = 4$
Also $2x_1 + c + 2x_2 + c = y_1 + y_2$ as B, D lie on y = 2x + c from above two equations c = –4.

$\therefore$ equation of BD is y = 2x – 4

Angle between CD and BD is $\frac{\pi}{4}$.

Let slope of CD is m, slope of BD is 2.

$\Rightarrow \tan\frac{\pi}{4} = \frac{2-m}{1+2m}$

$\Rightarrow$ 2 – m = 1 + 2m

$\Rightarrow m = \frac{1}{3}$

Equation of CD is $(y - 1) = \frac{1}{3}(x - 5)$

$\Rightarrow$ x – 3y – 2 = 0
Solving equations of BD and CD gives D(2, 0) from (1) C is (4, 4).
$\therefore$ Vertices of rectangle are A(1, 3), B(4, 4), C(5, 1) and D(2, 0).
Middle point of BC is (9/2,5/2).

15. 4 ABCD is a parallelogram whose diagonals are parallel to the lines 2y – x – 5 = 0 and y + 2x – 7 = 0 respectively.
The parallel lines to these lines are:
2y – x – k = 0

$\Rightarrow y = \frac{x}{2} + \frac{k}{2} = 0$ … (i)

and y + 2x – m = 0

$\Rightarrow y = -2x + \frac{m}{2}$ … (ii)

Slope of line (i) = $\frac{1}{2}$

Slope of line (ii) = – 2

Product of slopes = $\frac{1}{2} \times (-2) = -1$

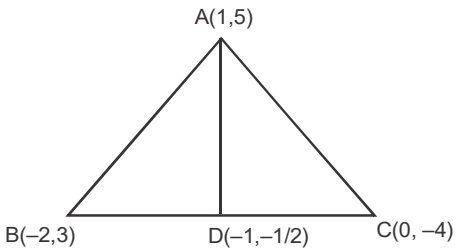
Therefore, lines (i) and (ii) are perpendicular to each other.
Since diagonals of a rhombus bisect each other at right angles. Hence, the parallelogram ABCD is a rhombus.

16. 2 Centroid divides orthocenter and circumcenter in the ratio of 2 : 1
The co-ordinates of orthocentre and the centroid of a triangle ABC are (–5, 7) and (5, 5)
Let the co-ordinate of circumcenter be (x, y)

$\frac{2x-5}{3} = 5$ and $\frac{2y+7}{3} = 5$

x = 10 and y = 4
Hence the circumcenter = (10, 4)

17. 4



Here AD is the median, so D is the midpoint of BC

$\therefore D\left(-1, -\frac{1}{2}\right)$

The equation of AD is

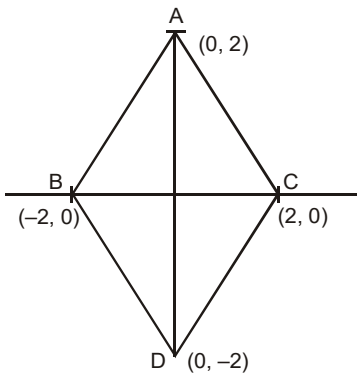
$$y - 5 = \frac{\frac{-1}{2} - 5}{-1-1}(x-1)$$

$$\Rightarrow y - 5 = \frac{11}{4}(x-1)$$

$\Rightarrow 4y - 20 = 11x - 11$

$\Rightarrow 11x - 4y + 9 = 0$

18. 3 The given curve is a rhombus as shown below



Area of rhombus = $\frac{1}{2}d_1 d_2$

AD = BC = 4

$\therefore$ The required area $= \frac{1}{2} \times 4 \times 4 = 8.$

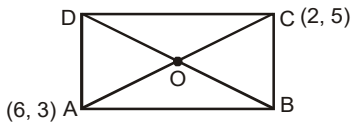
19. 1 $y = |x-1| + |x+1|$

at $x = \frac{1}{2}; y = \left|\frac{1}{2} - 1\right| + \left|\frac{1}{2}, 2\right| = \frac{1}{2} + \frac{3}{2} = 2$

Now, distance between $\left(\frac{1}{2}, 1\right)$ and $\left(\frac{1}{2}, 2\right)$

is $\sqrt{\left(\frac{1}{2} - \frac{1}{2}\right)^2 + (1-2)^2} = 1$

20. 4



Since diagonals of a rectangle bisect each other so O is the mid point of AC.

Co-ordinates of O $= \left(\frac{6+2}{2}, \frac{5+3}{2}\right)$ = (4, 4)

Point O will satisfy the equation of the other diagonal too. So,

y = 3x + c

4 = 3 × 4 + c

c = – 8.

21. 9 Equation of straight line = 3x + 5y – 45 = 0

It intersects 'X' axis when y = 0, $\Rightarrow$ at the point (15, 0)

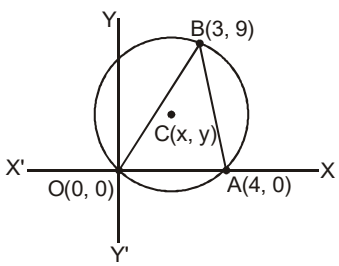
It intersects Y axis, when x = 0, $\Rightarrow$ at the point (0, 9)

Diameter of circumcircle = length of hypotenuse

$= \sqrt{15^2 + 9^2}$

$\therefore$ Radius $= \frac{\sqrt{225+81}}{2} \simeq 9$

22. 1



Let C(x, y) be the center of the circle.

OC = AC = BC = Radius

Then, $OC^2 = (x-0)^2 + (y-0)^2 = x^2 + y^2$ … (i)

$AC^2 = (x-4)^2 + (y-0)^2 = x^2 + 16 - 8x + y^2$ … (ii)

$BC^2 = (x-3)^2 + (y-9)^2 = x^2 + 9 - 6x + y^2 + 81 - 18x$ … (iii)

From (i) and (ii), x = 2

From (i) and (iii), y = 39/9

Therefore, radius = OC = $\sqrt{4+\left(\frac{39}{9}\right)^2}$

Hence, area of the circle

$= \pi \times OC^2 = \pi\left[4+\left(\frac{39}{9}\right)^2\right] = \frac{205\pi}{9}$ sq. units.

23. 4 The midpoint of (2, 1) and (–3, –4)

$= \frac{2-3}{2}, \frac{1-4}{2} = -\frac{1}{2}, -\frac{3}{2}$

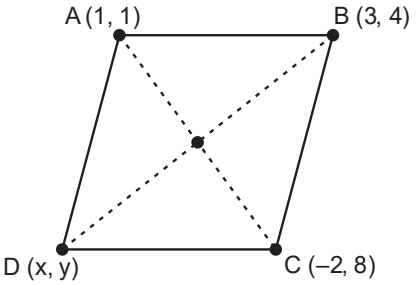
Since diagonals of a parallelogram bisect each other.

Therefore, $\left(-\frac{1}{2}, -\frac{3}{2}\right)$ lies on the line x + 9y + c = 0.

$-\frac{1}{2} + 9 \times \left(-\frac{3}{2}\right) + c = 0$

$\Rightarrow$ c = 14.

24. 1



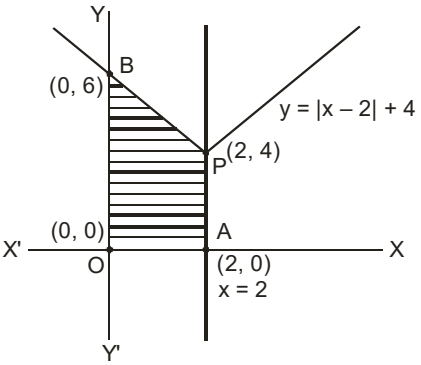
The mid points of AC and BD coincide.

$\Rightarrow \frac{1+(-2)}{2} = \frac{x+3}{2} \Rightarrow x = -4$

$\frac{(1+8)}{2} = \frac{y+4}{2} \Rightarrow y = 5$

Hence, co-ordinates of D are (–4, 5).

25. 4



Required area = Area of trapezium formed by the points (0, 0), (2, 0), (2, 4) and (0, 6)

Hence, area of trapezium = 1/2 × Height × Sum of parallel sides

= 1/2 × 2 × (6 + 4) = 10 sq. units.